Surface scattering in metallic nanoshells

This set of notes concerns the paper The limitation of electron mean free path in spherical nanosize particles with a metal shell (https://books.google.com/books?hl=en&lr=&id=JsPMi55EH-sC&oi=fnd&pg=PA103&dq=S.M.+Kachan,+A.N.+Ponyavina,+Reviews+and+Short+Notes+to+Nanomeeting%2799,+1999,+pp.103.&ots=ZdI68WeXzp&sig=Ty07vDFQFpUa71SbfcJUfT9pAJk#v=onepage&q&f=false)

In particular, we consider a nanoshell, with inner radius, r and outer radius, R. We wish to find the mean free path in such a system for the average electron. Thus, if the electron gas is uniformly distributed, we wish to find the quantity

$$\frac{r^2 L_r}{r^2+R^2}+\frac{R^2 L_R}{r^2+R^2},$$

where L_r is the average mean free path for an electron starting from the inner sphere and L_R is the average mean free path for an electron starting from the outer sphere. In calculating L_r and L_R, we assume the angle at which the electron leaves the surfaces is uniformly distributed, $0<\theta<\pi/2$, where the angle θ is measured from the normal of the surface.

Derivation of the individual mean free paths

Scattering from the outer surface to the inner surface

Using the law of cosines in conjunction with the law of sines, we obtain:

$$L^2=R^2+r^2-2Rr\cos(\pi-\phi-\theta)=R^2+r^2+2Rr\cos(\phi+\theta)= \\ R^2+r^2+2Rr(\cos(\theta)\cos(\phi)-\sin(\theta)\sin(\phi)),$$

where $\sin(\phi) = (R/r)\sin(\theta), \cos(\phi) = -\sqrt{1-(R/r)^2\sin^2(\theta)}$, where the sign for $\cos(\phi)$ was chosen by the fact that $\pi/2 < \phi < \pi$, meaning $\cos(\phi) < 0$. Using these two relations, we find:

$$L^2 = R^2 + r^2 + 2Rr\Bigg[-\cos(\theta)\sqrt{1-(R/r)^2\sin^2(\theta)} - (R/r)\sin^2(\theta)\Bigg] = \\ R^2\Big[\cos^2(\theta)-\sin^2(\theta)\Big] + r^2 - 2R\cos(\theta)\sqrt{r^2-R^2\sin^2(\theta)} =$$

At this point, we define $a \equiv R\cos(\theta)$ and $b \equiv \sqrt{r^2 - R^2\sin^2(\theta)}$ and we note that we may write:

$$L^2 = (a-b)^2 \rightarrow L = R\cos(\theta) - \sqrt{r^2-R^2\sin^2(\theta)}$$

Equivalently, we may write:

$$L = R\cos(\theta) - r\sqrt{1-(1/\alpha)^2\sin^2(\theta)},$$

where $\alpha \equiv r/R$. This is the first relation in **Equation 4** of the paper. Note that the expression for L above is clearly only valid for $\sin(\theta) < \alpha$. When $\theta > \arcsin(\alpha)$, the electron misses the inner sphere and scatters to the outer sphere.

We denote the integral of this mean free path by $I_1 \equiv I_1^{log} + I_1^0$, where the first term is given by:

$$I_1^{log} \equiv -r\int_0^{\arcsin(\alpha)} \sqrt{1-(1/\alpha)^2\sin^2(\theta)}\sin(\theta)\mathrm{d}\theta$$

and the second term is given by:

$$I_1^0 \equiv R\int_0^{\arcsin(\alpha)} \cos(\theta)\sin(\theta)\mathrm{d}\theta$$

Scattering from the outer surface to the outer surface

As mentioned above, when $\theta > \arcsin(\alpha)$, we scatter from the outer surface to the outer surface. In this case, the mean free path as a function of angle is given by (using the law of cosines):

$$L^2 = R^2 + R^2 - 2R^2\cos(\pi - 2\theta) = 2R^2(1 + \cos(2\theta)) = 4R^2\cos^2(\theta)$$

From this, we immediately find $L = 2R\cos(\theta)$

This is the second relation in **Equation 4** of the paper.

We denote the integral of this quantity as I_2^0, where we define the quantity I_2^0 as follows:

$$I_2^0 \equiv 2R\int_{\arcsin(\alpha)}^{\pi/2} \cos(\theta)\sin(\theta)\mathrm{d}\theta$$

Scattering from the inner surface to the outer surface

In this case, we have (using the law of cosines and sins):

$$L^2 = R^2 + r^2 - 2Rr\cos(\beta) \text{ with } \sin(\beta) = L\sin(\theta)/R \to \cos(\beta) = \\ = \sqrt{1 - (L/R)^2\sin^2(\theta)}$$

Therefore, we may write an equation with the only unknown variable being L:

$$L^2 = R^2 + r^2 - 2Rr\sqrt{1 - (L/R)^2\sin^2(\theta)}$$

At this point, we define $\alpha \equiv r/R$, $l \equiv L/R$, to write a less messy equation:

$$l^2 = 1 + \alpha^2 - 2\alpha\sqrt{1 - l^2\sin^2(\theta)} = \left[\alpha - \sqrt{1 - l^2\sin^2(\theta)}\right]^2 + l^2\sin^2(\theta) \to$$

$$l^2\cos^2(\theta) = \left[\alpha - \sqrt{1 - l^2\sin^2(\theta)}\right]^2 \to l\cos(\theta) = \pm\left[\alpha - \sqrt{1 - l^2\sin^2(\theta)}\right],$$

Therefore, by putting the square root on one side and squaring the relation above we have:

$$\begin{aligned} l\cos^2(\theta) + \alpha^2 \mp 2\alpha l\cos(\theta) = 1 - l^2\sin^2(\theta) \to \\ l^2 \mp 2\alpha l\cos(\theta) + \alpha^2 - 1 = 0 \to \\ l = \pm\alpha\cos(\theta) \pm \sqrt{\alpha^2\cos^2(\theta) - (\alpha^2 - 1)} = \pm\alpha\cos(\theta) + \sqrt{1 - \alpha^2\sin^2(\theta)} \to \\ l = -\alpha\cos(\theta) + \sqrt{1 - \alpha^2\sin^2(\theta)}, \end{aligned}$$

where in getting rid of the second $\pm$ sign, we used the fact that $\alpha^2\cos^2(\theta) = \alpha^2 - \alpha^2\sin^2(\theta) < 1 - \alpha^2\sin^2(\theta)$, which necessarily means $\alpha\cos(\theta) < \sqrt{1 - \alpha^2\sin^2(\theta)}$ and of course we require $l > 0$. In getting rid of the first $\pm$ sign, we note that there are, of course, two points of intersection between any line and a sphere. To find the mean free path, we naturally choose the point closest to the inner sphere.

Equivalently, we may write the mean free path in terms of the unnormalized length:

$$L = R\left[-\alpha\cos(\theta) + \sqrt{1 - \alpha^2\sin^2(\theta)} \right]$$

This is the third relation in **Equation 4** of the paper. We denote the integral of this mean free path with $I_3 \equiv I_3^{log} + I_3^0$, where the first term corresponds to the integral:

$$I_3^{log} \equiv \int_0^{\pi/2} R\sqrt{1 - \alpha^2\sin^2(\theta)}\sin(\theta)\mathrm{d}\theta$$

and the second term corresponds to the integral:

$$I_3^0 = -R\alpha\int_0^{\pi/2}\cos(\theta)\sin(\theta)\mathrm{d}\theta$$

Performing the integrals

We first perform one of the more difficult integrals, **corresponding to integrating the square root part of the third relation, i.e.** I_3^{log}:

$$I_3^{log} = R\int\sqrt{1-\alpha^2(1-\cos^2(\theta))}\sin(\theta)\mathrm{d}\theta = \\ = R\int\sqrt{(1-\alpha^2)+\alpha^2\cos^2(\theta)}\sin(\theta)\mathrm{d}\theta = \\ = -R\alpha\int\sqrt{(1-\alpha^2)/\alpha^2+x^2}\mathrm{d}x,$$

where the integration range is from $0<\theta<\pi/2 \leftrightarrow 0<\cos(\theta)\equiv x<1$. To evaluate this integral, we use the identity, **Equation 30** from Table of Integrals (https://www.physics.umd.edu/hep/drew/IntegralTable.pdf)

$$I_3^{log} = R\alpha\int_0^1\sqrt{(1-\alpha^2)/\alpha^2+x^2}\mathrm{d}x = \frac{R\alpha}{2}\Bigg[\sqrt{1+(1-\alpha^2)/\alpha^2}+ \\ (1-\alpha^2)/\alpha^2\ln\left[\frac{1+\sqrt{1+(1-\alpha^2)/\alpha^2}}{\sqrt{(1-\alpha^2)/\alpha^2}}\right]\Bigg] = \frac{R\alpha}{2}\left[\frac{1}{\alpha}+\frac{1-\alpha^2}{2\alpha^2}\ln\frac{1+\alpha}{1-\alpha}\right],$$

This quantity must be weighed with a factor to account for the ratio of the inner surface's area to the total surface area:

$$\frac{r^2}{r^2+R^2} = \frac{\alpha^2}{1+\alpha^2},$$

which gives us for the integral:

$$\frac{r^2 I_3^{log}}{r^2+R^2} = \frac{R}{2}\left[\frac{\alpha^2}{1+\alpha^2}+\frac{1-\alpha^2}{2(1+\alpha^2)}\alpha\ln\frac{1+\alpha}{1-\alpha}\right]$$

There is a second integral that has a similar form, which arises from scattering from the outer surface to the inner surface (**which corresponds to integrating the square root part of the first relation**, i.e. I_1^{log}). In this case, the integral is of the form as above, except with $\alpha\to 1/\alpha$ and the integration range of x spans 1 to $\sqrt{1-\alpha^2}$:

$$I_1^{log} \equiv -r\int_0^{\arcsin(\alpha)} \sqrt{1-(1/\alpha)^2\sin^2(\theta)}\sin(\theta)\mathrm{d}\theta =$$
$$-\frac{r}{\alpha}\int_{\sqrt{1-\alpha^2}}^1 \sqrt{x^2-(1-\alpha^2)}\mathrm{d}x = -\frac{r}{2\alpha}\Bigg[\sqrt{1-(1-\alpha^2)}-$$
$$(1-\alpha^2)\ln\frac{1+\sqrt{1-(1-\alpha^2)}}{\sqrt{1-\alpha^2}}\Bigg] = -\frac{r}{2\alpha}\left[\alpha-\frac{(1-\alpha^2)}{2}\ln\frac{1+\alpha}{1-\alpha}\right],$$

where we again used **Equation 30** from Table of Integrals (https://www.physics.umd.edu/hep/drew/IntegralTable.pdf). This quantity must be weighed with the factor

$$\frac{R^2}{r^2+R^2} = \frac{1}{1+\alpha^2},$$

which accounts for the ratio of the outer sphere's area to the total area of both spheres. Taking this into account, we obtain:

$$\frac{R^2 I_1^{log}}{r^2+R^2} = -\frac{R}{2}\left[\frac{\alpha}{1+\alpha^2}-\frac{1-\alpha^2}{2(1+\alpha^2)}\ln\frac{1+\alpha}{1-\alpha}\right]$$

At this point, we combine the two integrals thus computed to give us:

$$\frac{R^2 I_1^{log}+r^2 I_3^{log}}{r^2+R^2} = \frac{R}{2}\left[\frac{\alpha^2}{1+\alpha^2}-\frac{\alpha/2}{1+\alpha^2}+\frac{1-\alpha^2}{2(1+\alpha^2)}(1+\alpha)\ln\frac{1+\alpha}{1-\alpha}\right]$$

The other integrals combined give us:

$$\frac{R^2(I_1^0+I_2^0)+r^2I_3^0}{r^2+R^2} = \frac{R}{1+\alpha^2}\Bigg[\alpha^2\Bigg[-\alpha\int_0^{\pi/2}\sin(\theta)\cos(\theta)\mathrm{d}\theta\Bigg] + \Bigg[\int_0^{\arcsin(\alpha)}\sin(\theta)\cos(\theta)\mathrm{d}\theta + 2\int_{\arcsin(\alpha)}^{\pi/2}\sin(\theta)\cos(\theta)\mathrm{d}\theta\Bigg]\Bigg] =$$

$$\frac{R}{1+\alpha^2}\Bigg[\frac{\alpha^3}{4}(-2) - \frac{1}{4}\Bigg[\cos(2\arcsin(\alpha))-1\Bigg] - \frac{1}{2}\Bigg[-1-\cos(2\arcsin(\alpha))\Bigg]\Bigg]$$

We simplify this as:

$$\frac{R}{1+\alpha^2}\Bigg[-\alpha^3/2+\alpha^2/2+1-\alpha^2\Bigg] = \frac{R}{1+\alpha^2}\Bigg[-\alpha^3/2-\alpha^2/2+1\Bigg]$$

The final answer

Thus the mean free path is given by (with all terms included):

$$R\Bigg[-\alpha/2+\frac{1}{1+\alpha^2}+\frac{1-\alpha^2}{4(1+\alpha^2)}(1+\alpha)\ln\frac{1+\alpha}{1-\alpha}\Bigg]$$

Note that in our derivation we find a factor of $(1+\alpha)$ instead of $(1-\alpha)$ as a coefficient of the logarithmic term...

Attempting to see if final answer is correct

In the below, we define $r_1 \equiv r$ (the inner radius) and $r_2 \equiv R$ (the outer radius). We make these definitions for easier comparison with the paper cited at the end of this section.

We write the mean free path derived above as follows:

$$r_2\left[-\frac{r_1}{2r_2}+\frac{r_2^2}{r_2^2+r_1^2}-\frac{(1-r_1/r_2)(r_1+r_2)^2}{4(r_2^2+r_1^2)}\ln\frac{r_2-r_1}{r_2+r_1}\right]=$$

$$-\frac{r_1}{2}+\frac{r_2^3}{r_2^2+r_1^2}-\frac{(r_2-r_1)(r_1+r_2)^2}{4(r_2^2+r_1^2)}\ln\frac{r_2-r_1}{r_2+r_1}=$$

$$\frac{1}{4(r_2^2+r_1^2)}\left[-2r_1(r_1^2+r_2^2)+4r_2^3-(r_2-r_1)(r_1+r_2)^2\ln\frac{r_2-r_1}{r_2+r_1}\right]$$

Our result is consistent with Electron Mean Free Path in a Spherical Shell Geometry (https://pubs.acs.org/doi/pdf/10.1021/jp8010074?casa_token=QcyRQPJvruwAAAAA:surGr2EV4gA8_Uxt_AqPXBm-ypK2j3aGAN5819uorQPwz2_1Rr3R49_VRyvKN645v9dZDBEO1iqGEV0)

Thus, the first paper likely has a typo.